MIL-C-22896A(OS)
29 August 1968
Superseding
MIL-C-22896 (WE?)
25 June 1962

MILITARY SPECIFICATION

CONTACTOR

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1, SCOPE

1.1 Scope. This specification describes the requirements for the procurement of one type of contactor, which is to be used to connect and disconnect a 160 horsepower torpedo propulsion motor to a main battery source.

2. APPLICABLE DOCUMENTS

2,1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical
PPP-B-575	Box; Paper-Overbid Veneer (Strap-Around Type)
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Boxes; Wood, Wirebound
ppp-B-591	Boxes; Fiberboard, Wood-Cleated
PPP-B-601	Boxes; Wood, Cleated-Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
PPP-B-636	Boxes; Fiber
ppp-B-640	Boxes; Folding, Fiberboard, Corrugated, Triple Wall
PPP-B-645	Boxes; Folding, Fiberboard, Heavy Duty

MILITARY

MIL-P-116	Preservation, Methods of

FSC 1355

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MIL-E-17555	Electronics and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of
MIL-T-22361	Thread Compound, Antiseize, Zinc Dust Petrolatum
MIL-E-22403	Environmental Requirements and Tests (For Torpedo Mark 45 Mod 0 and Mod 1 Components and Assemblies)

STANDARDS

MILITARY

MIL-STD-125	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

Drawing 2209934	Contactor, Electrical Assembly

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements
NAVORD OD 12301	Insulation Resistance and High Potential Measurements and Procedures
NAVORD OD 12302	Procedure for Assembly and Adjustment of Contactor

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 General requirements

3.1.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the contactor shall be manufactured by the methods and procedures proposed for the production lot. These samples shall be tested as specified in Section 4 herein and are for the purpose of determining that the production items meet the requirements of this specification and the design. See 4.2.1.

3.1.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the contactor shall conform to the detailed requirements given on NOSC drawing 2209934.

3.1.3 Finishes, coating construction, and dimensions. Finishes, coatings, construction, and dimensions for the contactor shall be in accordance with NOSC drawing 2209934.

3.1.4 Moisture, fungus, and corrosion. The contactor shall be rendered moisture, fungus, and corrosion resistant in accordance with OS 6341.

3.1.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with MIL-STD-130, where applicable.

3.1.6 Interchangeability. Component parts of the assembled units (unless otherwise specified on the applicable drawings) fabricated in accordance with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required, then changes in dimensions are necessary, and revision and correction of the applicable drawings is required. See 3.1.9.

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3.1.7 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings, If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages,

3.1.8 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.1.9 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation clarification, and resolution (or correction),

3,2 Definitions. Definitions applicable to the environmental requirements of this specification are set forth in MIL-E-22403.

3,3 Performance requirements and product characteristics. The contactor shall meet the following performance requirements and product characteristics:

3.3.1 Coil circuit resistance. The resistance between pins A and C of J2301 shall be 10.5 ohms plus or minus 10 percent, when tested in accordance with 4.5.4.1.

3.3.2 Insulation resistance. The insulation resistance between the various termihals listed ih Table I shall be greater than 100 megohms, when tested in accordance with 4.5.4.2.

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Table I

Connections for Insulation Resistance Test

From		To
A	J-2301	Metal Housing
B	J-2301	Metal Housing
c	J-2301	Metal Housing
D	J-2301	Metal Housing
E	J-2301	Metal Housing
+	Main Terminal	Metal Housing
-	Main Teminal	Metal Housing

3.3.3 Dielectric strength. The contactor shall be capable of withstanding 1500 volts root-mean-square (rms), 60 cycles per second (cps), applied between any of the main contact terminal studs and the housing for 60 seconds; and 600 volts rms between any of the closing and tripping terminals and the metal housing for 60 seconds, when tested in accordance with 4.5.4,3.

3.3.4 Minimum closing voltage, The main contacts of the contactor shall close when a potential of 22 volts dc plus zero minus 0.5 volt is applied to the close-coil, when the contactor is wired and tested in accordance with 4.5.4.4.

3,3.5 Maximum holding voltage. With main contacts closed as in 3.3.4, the contacts shall remain closed when the close voltage is seduced to 5 plus zero minus 2 volts, when the contactor is tested in accordance with 4.5.4.4.1.

3.3.6 Closing time. When 25 volts dc is applied to the close-coil and a current of 2 amperes is passed through the main contacts, the close time shall not exceed 0,050 second, after which contact bounce shall have ended, when tested in accordance with 4.5.4.5

3.3,7 closing capacity. The contactor shall be capable of connecting a Dower supply having an open circuit voltage of 310 volts dc maximum and an operating voltage of 250 volts dc nominal across a load having a resistance of 0.44 ohm and an inductance of 0,003 henry, when tested in accordance with 4.5.4.6,

3.3.8 Temperature rise. The contactor shall be capable of carrying a current of 560 amperes plus or minus 25 amperes to the load specified in 3.3.7 for a period off 15 minutes, The temperature rise from ambient (65 degrees Fahrenheit (F) to 95 degrees F) shall not exceed 350 degrees F during the 15 minute period, when tested in accordance with 4.5.4.7.

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3.3.9 Voltage drop across the main contact, The voltage drop across the main contacts of the contactor shall not exceed 0.250 volt when the operating voltage is applied to the load specified in 3.3.7, with the maim contact in the closed position, when the contactor is tested in accordance with 4.5.4.8.

3.3.10 Interrupting capacity, The contactor shall be capable of interrupting the operating voltage specified in 3.3.7, when connected across the load specified in 3.3.7, when tested in accordance with 4.5.4.9

3.3.11 Sealing. The contactor assembly shall show no evidence of leakage when tested as specified in 4.5.4.10.

3.4 Environmental requirements. The contactor shall meet the other applicable requirements of this specification during or after tests defined or specified in 4,6.

3.4.1 Vibration operating condition. The contactor shall withstand the vibration requirements specified in MIL-E-22403, shall function during and after-vibration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.1,

3.4.2 Vibration, shipping condition. The contactor, packed and packaged in accordance with Section 5 herein, shall withstand the vibration requirements specified in MIL-E-22403, and thereafter **shall** show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.2.

3.4.3 Shock, nonoperating condition. The contactor shall withstand the shock requirements specified in MIL-E-22403, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.3.

3.4.4 Shock, shipping condition. The contactor, packed and packaged in accordance with Section 5 herein, shall withstand the shock requirements specified in MIL-E-22403, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.4,

3.4.5 Acceleration. The contactor shall withstand the acceleration requirements specified in MIL-E-22403, shall function during and after acceleration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.5.

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3.4.6 Temperature. The contactor shall withstand the temperature conditions specified in MIL-E-22403, shall function during and after the temperature tests, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification, See 4.6.6.

3.4.7 Humidity. The contactor shall withstand the humidity conditions speci ied in MIL-E-22403, and thereafter shall show *no* corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.7.

3.4.8 Tropical environment, The contactor shall withstand the tropical environment conditions specified in MIL-E-22403, and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification. See 4.6.8.

3.4.9 Marine atmosphere environment. The contactor shall withstand simulated sea atmosphere conditions and tests specified in MIL-E-22403, for salt spray testing, and thereafter shall show no evidence of corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.6.9.

3,4.10 Operating life. The operating life of the contactor shall be 100 cycles. One cycle consists of performing the test specified in 4.5.4.7 and allowing the load current to flow for one minute, then performing the test specified in 4.5.4.10 and allowing the unit to cool to ambient temperature. The permissible repairs to the contactor are specified in 3.4.11 and tests are as specified in 4.6.10.

3.4.11 Overhaul period. The contactor shall be capable of completing 50 cycles such as specified in 3.4.10 without repairs or replacement of parts. At the completion of the 50 cycles, the main contacts and arc chambers may be cleaned and any burned spots on the main contacts removed.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter, Burrs and sharp edges, as well as rosin flash, shall be removed.

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3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities *or* any commercial laboratory acceptable to the Government, The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified by the contract or purchase order, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with 4.2.1 and 4.2.3.

4.2.1 Preproduction sample. Unless otherwise specified for each contract or order, a representative sample of two contractors shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for preproduction tests at an activity designated by the Naval Ordnance Systems Command, for purposes of determining compliance with the requirements of the contract, specification and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government

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4,2.2 Lot size. As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot," i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise determined by the Government inspector, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. Where a shipment consists of less than 60 units the Government inspector may select and combine several shipments into one "inspection lot." In any event, all units of a single "inspection lot" shall be made to the same design, with the same material and manufacturing techniques, and shall be part of one continuous production run.

4.2.3 Periodic production samples. The Government inspector shall randomly select from each production lot a number of units equal to 10 percent of the lot size, but not less than 2 nor more than 40 units. Unless otherwise specified in the contract or purchase order, these units shall be submitted to a Naval Ordnance Systems Command assigned activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The contractors shall be considered satisfactory if they meet all specifications and pertinent drawing requirements and withstand the tests specified in 4.5 and 4.6. Periodic production samples accepted will be applied as part of the quantity specified by the contact or order, but shall not be used in any equipment manufactured for the Government.

4.3 Classification of tests. The inspection and testing of the contactor shall be classified as follows:

4.3.1 Quality conformance tests. These tests, detailed in 4.5 shall be accomplished by the contractor on each contactor submitted for' acceptance under contract.

4.3.2 Preproduction and periodic production tests. Preproduction and periodic production tests shall be accomplished on samples selected as specified in 4.2.1 and 4.2.3, which are representative of the production of the contractors after the award of the contract, to determine that the production meets the requirements of this specification. These tests are detailed in 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot or the cessation of production, as determined by the procuring activity.

4.3.3 Tests after delivery. Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein, as deemed necessary, after delivery of the contactor. Any defects discovered during or after these tests due to the contractor's failure to comply with the drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

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4.4 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 Power supplies

4.4.1.1 A dc power supply in which the voltage is variable from zero to 28 volts, under a load of at least 10 amperes, with a setting accuracy of plus or minus 0.5 volt.

4.4.1.2 A dc power supply in which the voltage is 250 volts plus or minus 10 percent, when delivering a current of 600 amperes plus or minus 20 amperes. Recovery shall be no more than 310 volts at zero amperes. The inductance of the power supply shall not exceed120 microhenries,

4.4.2 Voltmeters

4.4.2.1 DC voltmeter: Zero to 500 volt scale, 1000 ohms per volt, with an accuracy of plus or minus 5 percent of full scale.

4.4.2.2 DC voltmeter: Zero to 50 volt scale, 1000 ohms per volt, with an accuracy of plus or minus 5 percent of full scale.

4.4.2.3 DC vacuum tube voltmeter (VTVM): Zero to 3 volts, with an accuracy of plus or minus 3 percent of full scale, input resistance of 1 megohm.

4.4.3 DC ammeter: Zero to 1000 ampere scale plus or minus five percent. The shunt shall have an IR drop of 50 millivolts when the ammeter reads full scale.

4.4.4 Load rack: Variable from zero to 2,5 ohms, 140 kilowatts.

4.4.4.1 Duty cycle: 15 minutes on, 30 minutes off; variation in resistance to be no greater than 10 percent during 15 minutes of operation under load.

4.4.5 Inductor load: 0.003 henry, capable of carrying 560 amperes continuously. The allowable resistance of the inductor shall be no greater than O.OO5 ohm.

4.4.6 Calibrated oscilloscope: Tektronix type 535, or equivalent.

4.4.6.1 Range of measurements: 250 volts dc, full scale width, one to ten multiplier.

4.4.6.2 Sweep: Recurrent, 5 cps to 100 kilocycles.

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4.4.6.3 Frequency response: Down not more than two decibels (dbs) at 100 kilocycles (kc).

4.4.7 Switches: Single-pole, single-throw, capable of handling 10 amperes and 28 volts.

4.4.7.1 Circuit breaker: 600 amperes at 250 volts dc with overload trip not to exceed 125 percent of rated current and with remotely operated electromagnetic trip.

4.4.8 Temperature recorder: Minneapolis Honeywell Model 153X62-P6, or equivalent. The recorder must be capable of recording at least two points simultaneously, shall have a range from zero to 500 degrees F, and an accuracy of plus or minus two percent. It shall have a response time no greater than 100 milliseconds.

4.4.8.1 Sensing element: Copper Constantan Thermocouple, with a response time no greater than two seconds then mounted on the terminal board. The thermocouple shall be electrically insulated from the terminal, and securely mounted and insulated from the terminal with an insulating epoxy.

4.4.9 High potential test set: The high potential test set shall be continuously variable and capable of furnishing from zero to 2000 volts ac, at 60 cps, It shall have a sensing element to indicate insulation failure and a meter to indicate terminal voltage.

4.4.10 Ohmmeter: Capable of indicating resistances from zero to one megohm, with an accuracy of plus or minus five percent,

4.4.11 Megohmmeter: Capable of measuring insulation resistance at 500 volts dc over a range of from zero to 20,000 megohms, with an accuracy of plus or minus five percent at 20,000 megohms.

4.5 Quality conformance tests

4.5.1 The contractor shall conduct quality conformance tests as specified in 4.3.1 to ensure that the contactor is in compliance with the requirements of 3.1, 3.3, 3.5, and 3.6.

4.5.2 Test conditions. Unless otherwise specified, the contactor shall be subjected to acceptance tests under the following conditions:

4.5.2.1 Temperature: Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18 degrees centigrade (C) to 35 degrees C),

4.5.2.2 Altitude: Normal ground atmospheric pressure.

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4.5.2.3 Vibration: None.

4.5.2.4 Humidity: Room ambient to 95 percent relative maximum,

4.5.3 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required accecptance tests.

4.5.4 Test procedure

4.5.4.1 Coil circuit resistance. Place ohmmeter leads across pins A and C of J2301. The resistance shall be 10.5 ohms plus or minus 10 percent, to comply with 3.3.1.

4.5.4.2 Insulation resistance. The resistance of the insulation shall be tested in accordance with the procedure specified in OD 12301. Connect the probes of the megohmmeter and check the insulation resistance between the terminals listed-in Table I. The resistances must be within the limits specified in 3.3.2.

4.5.4.3 Dielectric test. Adjust the Hi-potential test voltage specified in 4.4.9 to 1500 volts rms, 60 cps. Apply 1500 volts, 60 CDS, between any one of the main terminal studs and the metal housing for a period of 60 seconds, No arcing or insulation breakdown shall occur, for the contractor to meet the requirements of 3.3.3. Repeat the test, using the other main terminal stud.

4.5.4.3.1 Adjust the Hi-potential test voltage of 600 rms, 60 cps. Apply the test voltage between all of the pins of connector J2301 joined together and the metal housing for 60 seconds, after which no arcing or insulation breakdown shall occur, for the contactor to meet the requirements of 3.3.3.

4.5.4.4 Minimum closing voltage. Wire the contactor as shown in Figure 1, and adjust the power supply specified in 4.4.1.1 to 22 volts. Place SW-1 in the “on” position. The ohmmeter connected across the main contact terminals shall indicate continuity (closed contacts) to meet the requirements of 3.3.4. Keep SW-1 in the “on” position and perform test 4.5.4,4.1.

4.5.4.4.1 Maximum holding voltage, With the contactor wired as in 4.5.4.4, reduce the close coil voltage to 5 plus zero minus 2 volts. The main contacts must remain closed to meet the requirements of 3.3.5.

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4.5.4.5 Closing time. Wire the contactor as shown in Figure 2, and adjust R-1 to 10 ohms. Adjust oscilloscope as follows: Adjust vertical gain on the oscilloscope to 5 volts per centimeter; adjust horizontal sweep (time base) to 0.010 second per centimeter; connect vertical input to oscilloscope as shown in Figure 2; set oscilloscope trigger mode to “external trigger"; connect horizontal trigger to trigger input as shown in Figure 2 and apply 25 volts to “close” coil and load by placing SW-1 in the “on” position. The oscilloscope trace shall indicate that the main contacts have closed; i.e.,- the trace shall move from 25 volts (5 centimeters) to zero in the time specified in 3.3.6.

4.5.4.6 Closing capacity. Wire the contactor as shown in Figure 3. Adjust the load rack to 0.44 ohm and place the back-up breaker or switch in the “on” position. Place switch SW-1 in the "on" position. The current through the load and ammeter shall be 560 plus or minus 25 amperes when the line voltage is adjusted to 250 volts, for the contactor to meet the requirements of 3.3.7.

4.5.4.7 Temperature rise. Attach the thermocouple specified in 4.4.8.1 to the main contact terminals, taking care to insulate the element electrically with a piece of 0.01 thick mica securely held in place with an epoxy resin. Connect the thermocouple leads to the temperature recorder specified in 4.4.8. With the contactor wired and the circuit adjusted as specified in 4.5.4.6, allow a current of 560 plus or minus 25 amperes to flow though the-load and main contacts for 15 minutes. Continuously record the temperature rise of the main contact terminals during the 15 minute period. After 5 minutes of operations perform the test specified in 4.5.4.8. The temperature rise above ambient (65 degrees F to 95 degrees F) after 15 minutes of operation shall meet the requirements specified in 3.3.8.

4.5.4.8 Voltage drop. With the circuit wired and adjusted as specified in 4.5.4.7 and with the main contacts in the closed position, connect the voltmeter (4.4.2.3) across the main contact teminals of the contactor. The voltage drop across the main contacts, with 560 plus or minus 25 amperes flowing through the contacts, shall meet the requirement of 3.3.9.

CAUTION: To avoid damaging the voltmeter, do not connect it across the main contact terminals when contacts are open.

4.5.4.9 Interrupting capacity. At the conclusion of the tests specified in 4.5.4.7 and 4.5 .4.8 and with the contactor wired and adjusted as in 4.5.4.6, place SW-1 of Figure 3 in the “off” position. The main contacts must open (the load current indicated by the ammeter must drop to zero) to meet the requirements of 3.3.10.

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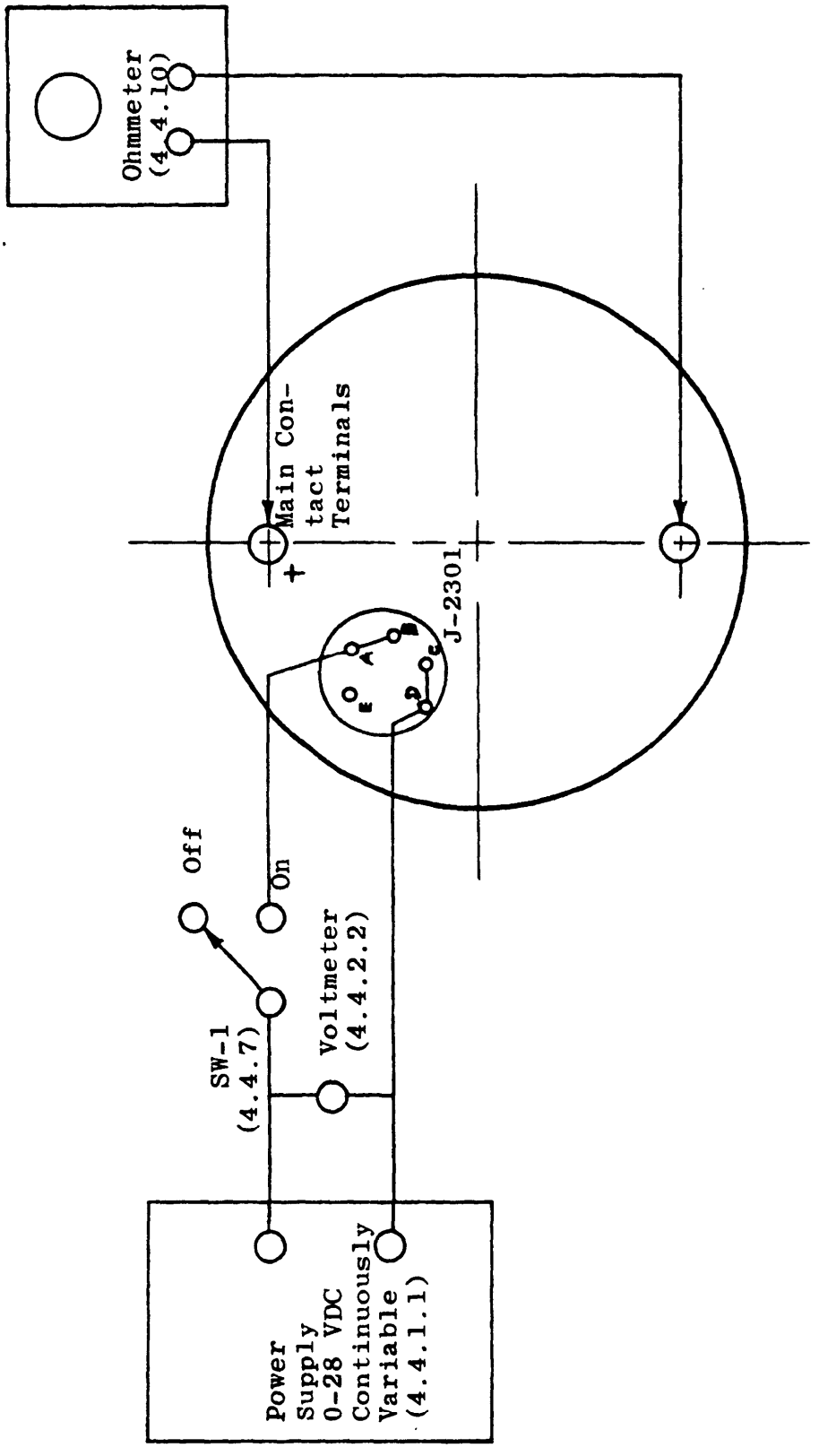


FIGURE 1. CLOSE VOLTAGE TEST SETUP

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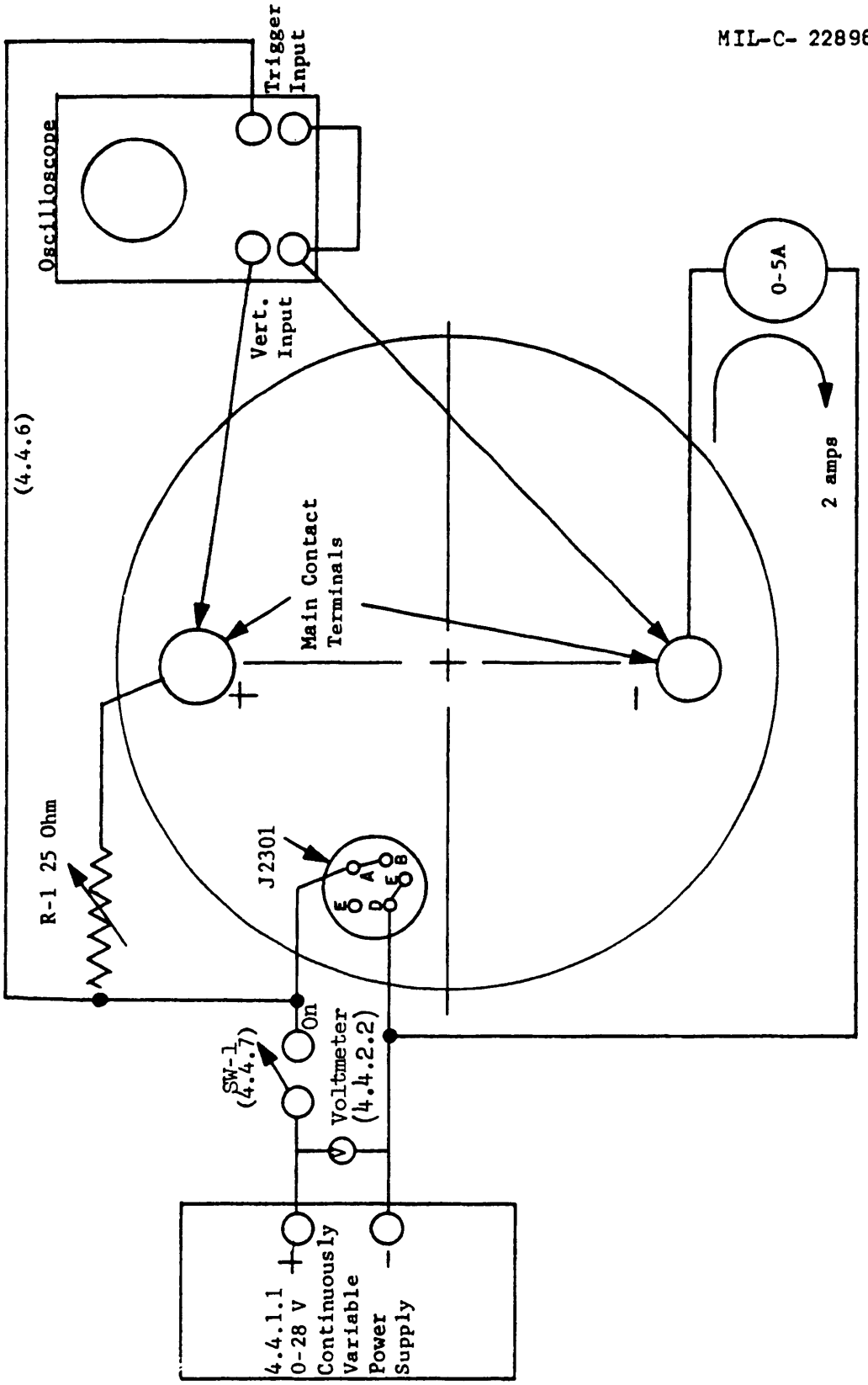


FIGURE 2. CLOSING TIME TEST SET UP

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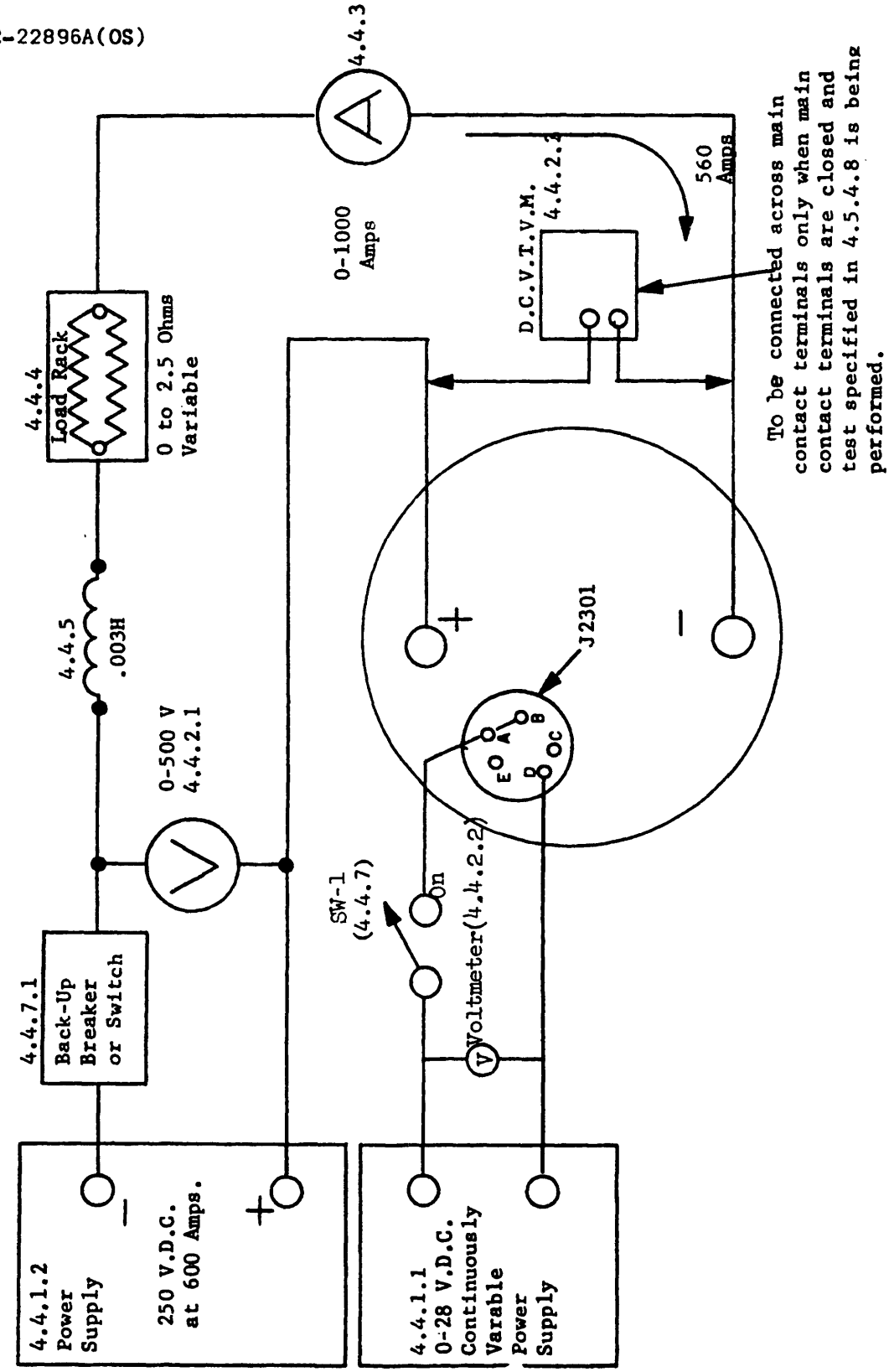


FIGURE 3. LOAD TEST

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4.5.4.10 Evacuation and sealing. Evacuate the contactor assembly through the 1/8 NPT filling port to a maximum of 25 and a minimum of 20 inches of mercury vacuum below atmospheric pressure. From the vacuum condition fill the assembly with nitrogen, BB-N-411, Type I, Class 1, Grade A or B, to a positive pressure of 8 plus 2 minus O psig. Disconnect the assembly, with gage and valve still attached, from the vac/fill equipment and immerse the assembly in a water bath containing a suitable wetting agent, With the assembly submerged for at least ten minutes, there shall be no bubbles and/or pressure drop caused by leakage, to meet the requirements of 3.3.11.

> CAUTION: Under no condition allow the assembly to remain submerged in water with less than two psig positive pressure inside the assembly. If no leakage is observed in ten minutes, remove assembly from water, dry it off, and, using the valve, stabilize pressure in assembly to two psig nominal. Remove the gage equipnent and promptly install plug AN933-C1 to minimize the amount of gas escaping from the unit. Prior to installing the plug, apply a thin coat of antiseize compound, MIL-T-22361, to its threads. Remove the excess compound after assembly,

4.6 Preproduction and periodic production tests

4.6.1 Vibration, operating condition. The contactor, rigidly mounted as shown in Figures 4 and 5, shall be subjected to vibration tests along the X, Y, and Z axes so as to meet the requirements specified in 3,4.1, and during and thereafter shall meet all the performance requirements and product characteristics of this specification.

4,6.2 Vibration, shipping condition. The contactor, rigidly mounted as shown in Figures 4 and 5, shall be vibrated along the X, ‘, and Z axes so as to meet the requirements specified in 3.4.2, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6,3 Shock, nonoperating condition. The contactor, rigidly mounted as shown in Figures 4 and 5, shall be subjected to shock tests along the X, Y, and Z axes so as to meet the requirements specified in 3.4.3, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6,4 Shock, shipping condition. The contactor, rigidly mounted as shown in Figures 4 and 5, shall be vibrated along the X, Y, and Z axes so as to meet the requirements specified in 3.4.4, and thereafter shall meet all the performance requirements and product characteristics of this specification.

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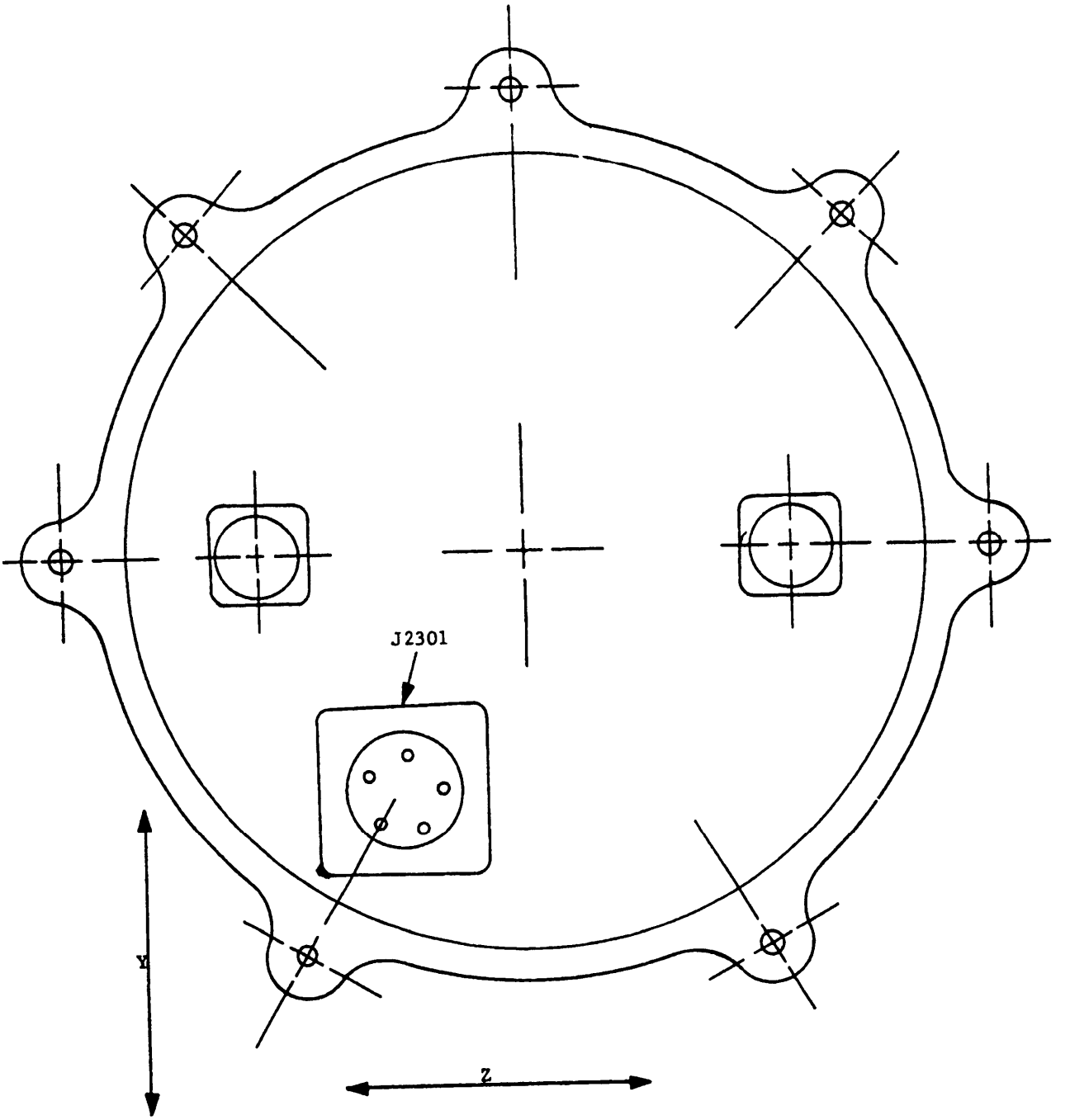


FIGURE 4. SHOCK AND VIBRATION AXES (Y AND Z AXES)

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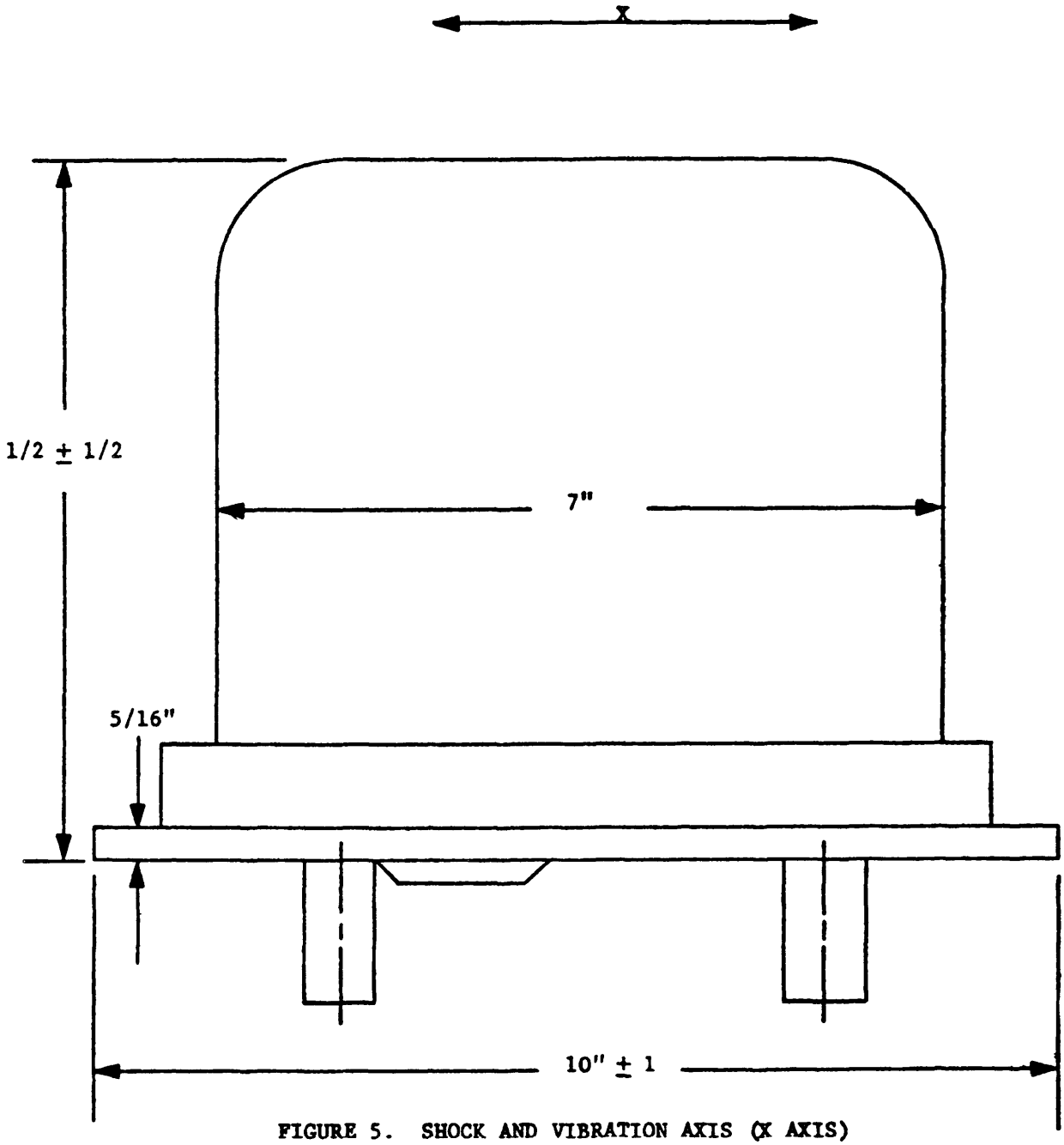


FIGURE 5. SHOCK AND VIBRATION AXIS (X AXIS)

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4.6.5 Acceleration. The contactor, rigidly mounted as shown in Figures 4 and 5, shall be subjected to acceleration tests along the X, Y, and Z axes so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6.6 Temperature. The contactor shall be subjected to temperature tests so as to meet the requirements specified in 3.4.6, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6.7 Humidity. The contactor shall be subjected to humidity tests so as to meet the requirements specified in 3.4.7, and thereafter shall meet all the performance requirements and product characteristics of this specifications

4.6.8 Tropical environment. The contactor shall be exposed to fungus tests so as to meet the requirements specified in 3.4.8, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6.9 Marine atmosphere environment. The contactor shall be exposed to simulated sea atmosphere tests so as to meet the requirements specified in 3.4.9, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6.10 Operating life, The contactor shall be wired as shown in Figure 3. The current through the load is adjusted to 560 plus or minus 25 amperes when the line voltage is adjusted to 250 volts. The power supply of 4.4.1.1 shall be set for 28 volts. The contactor shall be operated for 100 cycles as follows: Switch SW-1 shall be placed in the “on” position and the load current shall be allowed to flow through the main contacts for one minute; the load will be removed by placing switch SW-1 in the “off” position. The contactor shall be allowed to cool to the ambient temperature specified in 4.5.2.1; the contacts shall be cleaned after every 50 cycles. During the life cycle tests, the contactor shall meet all the performance requirements and product characteristics of this specification, without any replacement of parts, to meet the requirements of 3.4.10.

4.7 Rejection criteria. Failure of the contactor to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY ‘

5.1 Preservation and packaging

5.1.1 Level A

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5.1.1.1 Cleaning. Unless otherwise specified, the contactor shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the contactor shall be dried by either or both of Methods D-4 and D-1 of MIL-P-116.

5.1.1.3 Preservation application. Unless otherwise specified, no preservation shall be applied to the contactor.

5.1.1.4 Unit packaging. The contactor shall individually be packaged in accordance with Method IIc or IIe of MIL-P-116

Table II

(List of remissible shipping containers)

SPECIFICATION	DESCRIPTION	OVERSEAS MAX. WT. (LBS)	LEVEL A MAX. DIM. (INCHES)	DOMESTIC MAX. WT. (LBS)	LEVEL C MAX. DIM. (INCHES)
PPP-B-575	Box; Paper Overlaid Veneer (Strap-Around Type)	200	L+W+D=84	200	Same as Level .4
PPP-B-576	Box; Wood, Cleated, Veneer, Paper, Overlaid	200	any	500	any
PPP-B-585	Boxes; Wood, Wirebound	200	any	500	any
PPP-B-591	Boxes; Fiberboard, Wood-Cleated	200	any	400	any
PPP-B-601	Boxes, Wood, Cleated-Plywood	200	any	1000	any
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner	200	any	600	any
PPP-B-636	Boxes; Fiber	70	L+W+D=70*	160	L+W+D=120*
PPP-B-640	Boxes; Folding, Fiberboard Corrugated, Triple Wall	---	----	250	W+D=72 L+W=90
PPP-B-645	Boxes, Folding, Fiberboard, Heavy Duty	125	L+W+D=6C	125	Same as Level 4

*These are maximum dimensions for maximum weight. Lower weights have lower dimension maxima. See Specification.

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5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning, drying, and preservation. Cleaning, drying, and preservation of the contactor shall be in accordance with the manufacturer's commercial practice, when adequate,

5.1.3.2 Unit packaging. The contactor shall be unit packaged in accordance with Method III of MIL-P-116.

5.2 Packing

5.2.1 Number of packages per pack. The number of unit packages per pack shall be as specified in the contract or purchase order.

5.2.2 Level A. In addition to the following, Level A packing shall be in accordance MIL-E-17555.

5.2.2.1 Exterior containers. The contactor, packaged as specified in 5.1.1, shall be packed in any overseas type container given in Table II constructed in accordance with its applicable specification, provided the gross weight Or dimension, as given for that container, is not exceeded.

5.2.3 Level B. Level B packing shall be the same as Level A, except that exterior containers shall conform to the domestic type containers given in Table II.

5.2.4 Level C. The contactor shall be packed to afford protection against damage during direct shipnent from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation,

5.3 Marking

5.3.1 Normal markings. Unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-1290

5.3.2 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The contactor is intended for use in Torpedo Mk 45 Mod 0 and Mk 45 Mod 1 or other suitable application.

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6.2 Ordering data. Procurement documents should specify the specific title, number, and date of this specification, and exceptions to this specification; applicable drawings and other documents; selection of applicable levels of packing; and guarantee clause (see 6.5).

6.2.1 The attention of the contracting officer is called to 5.2.1.

6.3 Criteria for use of the proper level of packaging shall be as follows:

6.3.1 Level A. This level shall be used for all items except those receiving Level C packaging in accordance with 6.3.2.

6.3.2 Level C. This level shall be used for those items which are to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Criteria for use of the proper level of packing shall be as follows:

6.4.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.4.2 Level C. This level shall be used for those items which are to be shipped-to domestic installations for immediate use or repacking at the first receiving activity,

6.5 Guarantee. The contract or order shall include a requirement for a guarantee that any contactor which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor’s failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.6 The contract or order shall include a sufficient number of extra items to cover production and preproduction testing samples.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-22896A(OS) CONTACTOR

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426

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